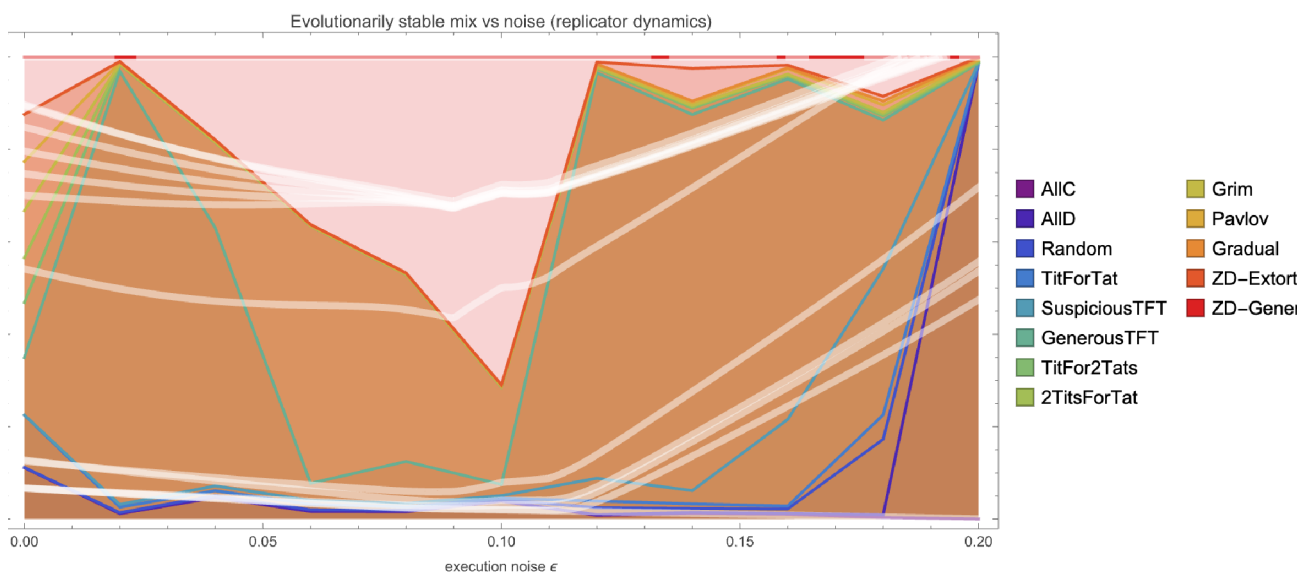


Re-running Axelrod's Tournament: Cooperation, Evolution, and Noise

The iterated Prisoner's Dilemma in the Wolfram Language – from Tit-for-Tat to an evolutionarily stable mix that depends on noise

Wolfram Community submission – May 2026



The headline result. As execution noise ϵ rises (left to right), the evolutionarily stable composition of strategies shifts – from a coexistence of generous reciprocators to a collapse into unconditional defection.

Abstract

In 1980 Robert Axelrod ran a now-famous computer tournament for the **iterated Prisoner's Dilemma**. The winner was the simplest entrant of all – **Tit-for-Tat**: cooperate first, then copy your opponent. This notebook reproduces that tournament in the Wolfram Language, with the classic strategies plus modern entrants (Generous Tit-for-Tat, Pavlov, Gradual, and Press – Dyson zero-determinant strategies).

It then does two things Axelrod's setup hints at but does not show. **First**, an *evolutionary* tournament: we let populations of strategies reproduce in proportion to their success and ask which mixture is stable. **Second**, we add *noise* – a small probability that an intended move is mis-executed – and show that the evolutionarily stable mixture **depends on the**

noise level. Finally, in the spirit of an automated-research loop, we let a search algorithm *discover* memory-one strategies that beat Tit-for-Tat under noise.

1. The Prisoner's Dilemma, and why iteration changes everything

Two players each choose, simultaneously, to **cooperate** (C) or **defect** (D). The payoffs satisfy $T > R > P > S$ and $2R > T + S$. With the classic Axelrod values $T=5, R=3, P=1, S=0$:

	opp C	opp D
I C	$R=3$	$S=0$
I D	$T=5$	$P=1$

In a *single* game, defection strictly dominates: whatever the opponent does, you score more by defecting. Yet mutual cooperation ($R=3$ each) beats mutual defection ($P=1$ each). That is the dilemma. Axelrod's insight was that when the game is **repeated**, the shadow of the future can sustain cooperation – a strategy can reward cooperation and punish defection in later rounds.

Encoding strategies in the Wolfram Language

We encode $C = 1$ and $D = 0$. A strategy is a pure function of the two histories of actual moves; the smallest interesting family is the **memory-one** strategies, parameterised by the probability of cooperating after each of the four possible previous outcomes:

```
memOne[open_, {pCC_, pCD_, pDC_, pDD_}] := Function[{me, opp},
  If[me === {},
    If[RandomReal[] < open, 1, 0],
    If[RandomReal[] < {pCC, pCD, pDC, pDD}[[stateIndex[Last[me], Last[opp]]]], 1, 0]]];

(* Tit-for-Tat: cooperate first, then copy the opponent's last move *)
TitForTat = memOne[1, {1, 0, 1, 0}];
Pavlov = memOne[1, {1, 0, 0, 1}]; (* win-stay, lose-shift *)
GenerousTFT = memOne[1, {1, 1/3, 1, 1/3}];
```

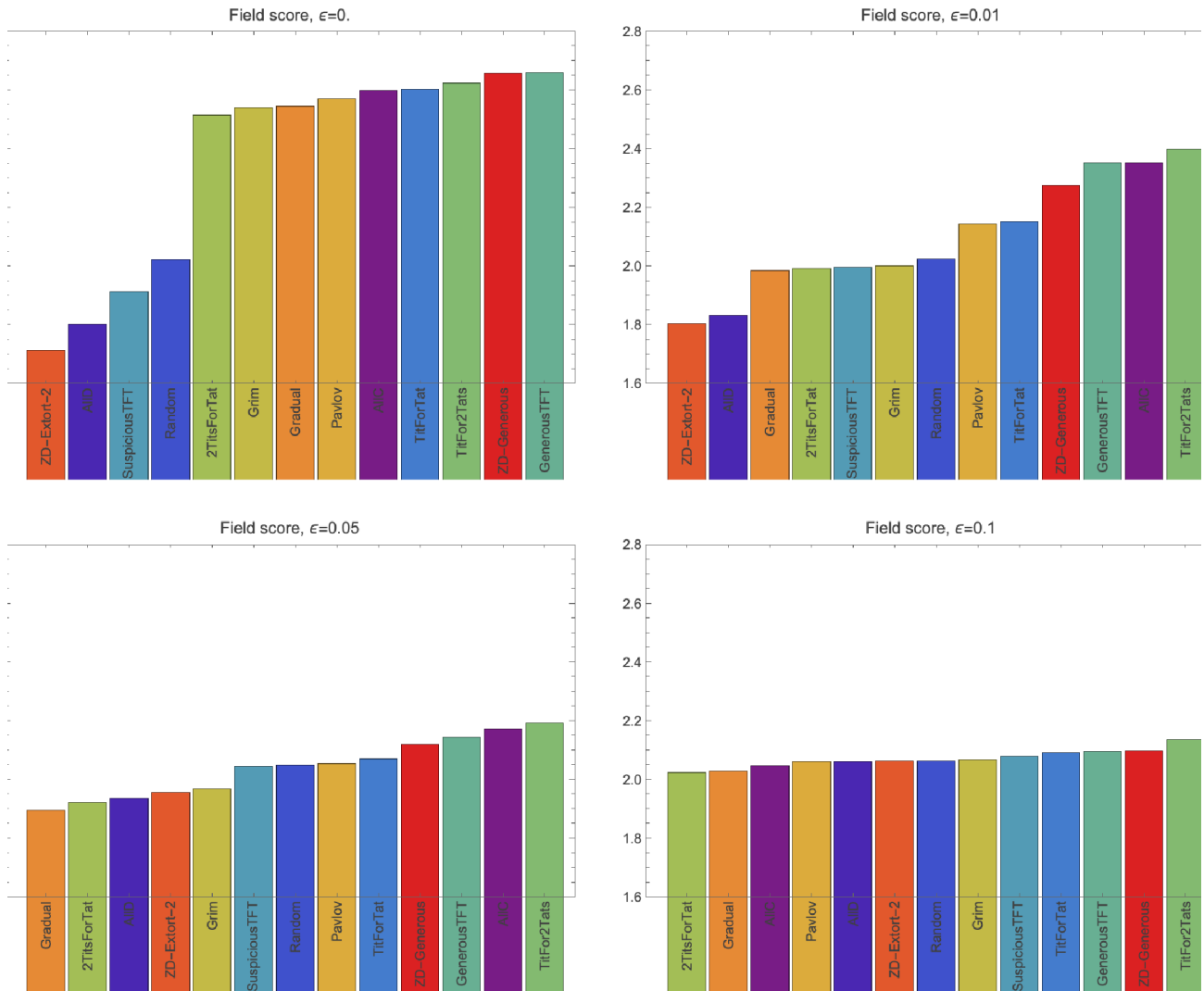
The full library (in `wolfram/axelrod.wl`) has 13 strategies: AllC, AllD, Random, Tit-for-Tat, Suspicious TFT, Generous TFT, Tit-for-2-Tats, 2-Tits-for-Tat, Grim, Pavlov, Gradual, and two zero-determinant strategies (an extortioner and a generous one).

Noise: the trembling hand

Real interactions are imperfect. We model **execution noise**: each intended move is flipped with probability ϵ , and the opponent sees the *actual* (post-noise) move. This single knob turns out to control everything that follows.

2. The classic round-robin (reproducing Axelrod 1980)

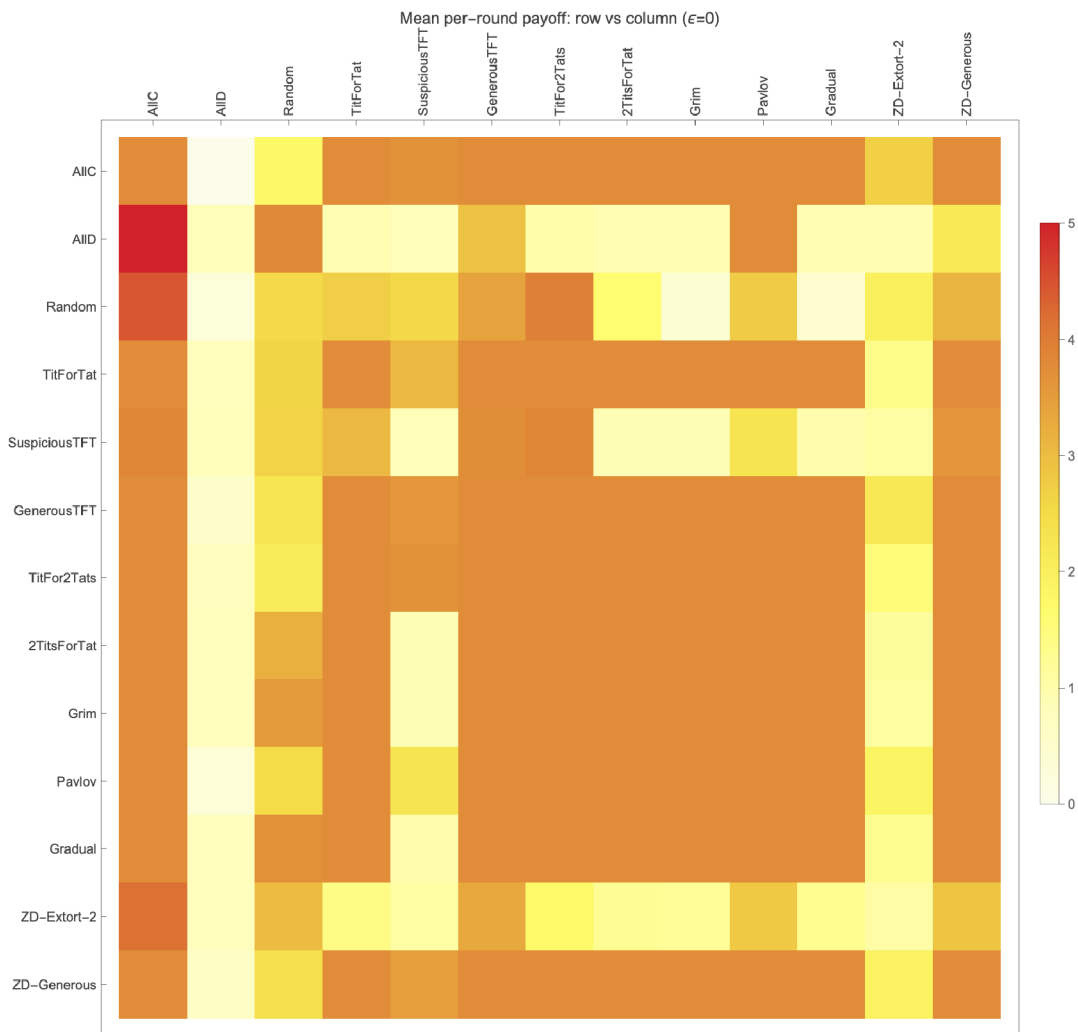
Every strategy plays every strategy (including itself) for 200 rounds, repeated 100 times to average out randomness. Each strategy's score is its mean per-round payoff against the whole field. At $\epsilon = 0$ (no noise) the ranking is:



Field score of each strategy at four noise levels. At $\epsilon=0$ the generous reciprocators lead; the extortioner (ZD-Extort-2) finishes last.

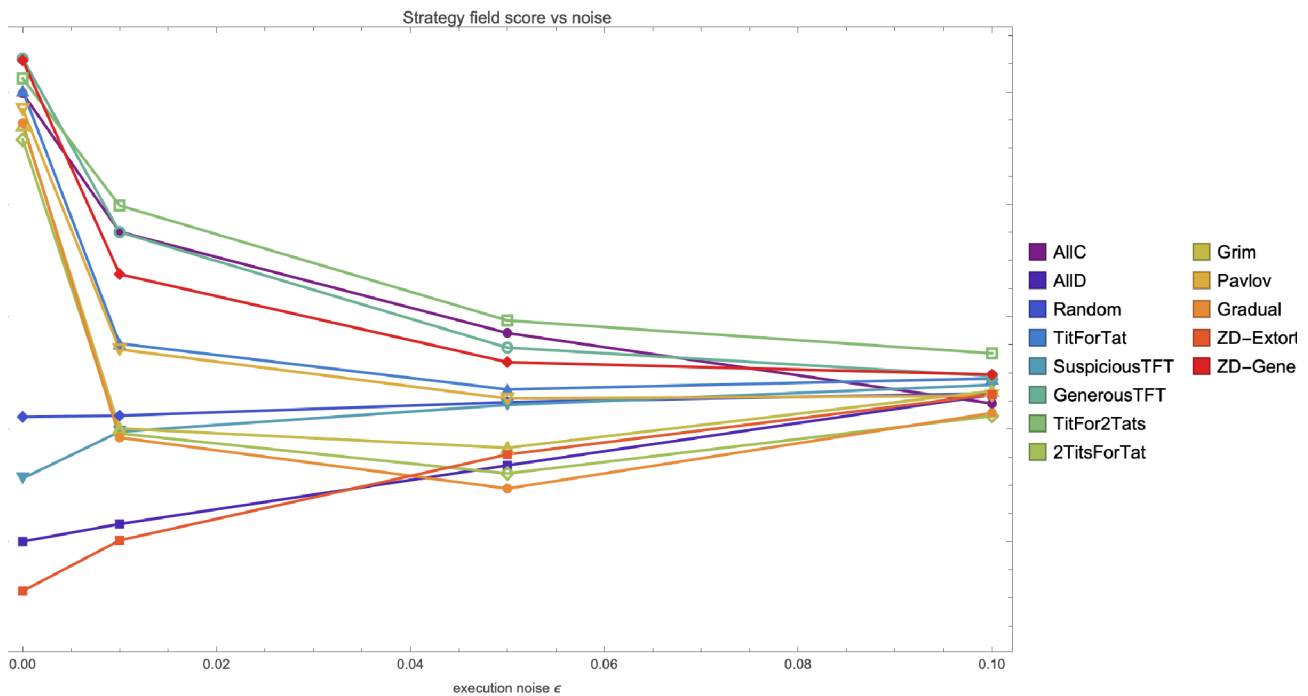
The classic lesson holds. Nice, reciprocal, forgiving strategies dominate. Notably, plain Tit-for-Tat is *edged out* by its more forgiving cousins – Generous TFT and Tit-for-2-Tats – exactly the refinement Axelrod found in his second tournament. And the zero-determinant **extortioner finishes dead last**: extortion wins individual matches but poisons the cooperative environment it depends on.

The payoff structure of who-beats-whom:



Mean per-round payoff, row strategy against column strategy ($\epsilon=0$). Warm = high. The cooperative cluster (top-left block) sustains mutual $R=3$; AllD and the extortioner do well in single matchups but drag the whole field down.

What noise does to the ranking



As noise rises, every strategy's score is pulled toward the mutual-defection payoff $P=1$, and the gap between 'nice' and 'nasty' strategies narrows. Unforgiving strategies (Grim, Gradual) fall fastest: a single accidental defection traps them in retaliation.

3. The evolutionary tournament – the stable mix depends on noise

Axelrod's deeper question was evolutionary: if successful strategies reproduce, which ones survive? We answer it two ways.

Replicator dynamics

Let x be the vector of population fractions and A the mean-payoff matrix ($A[i,j]$ = score of i against j). Each strategy grows in proportion to its fitness relative to the population average:

$$x_i' = x_i ((A \cdot x)_i - x \cdot A \cdot x)$$

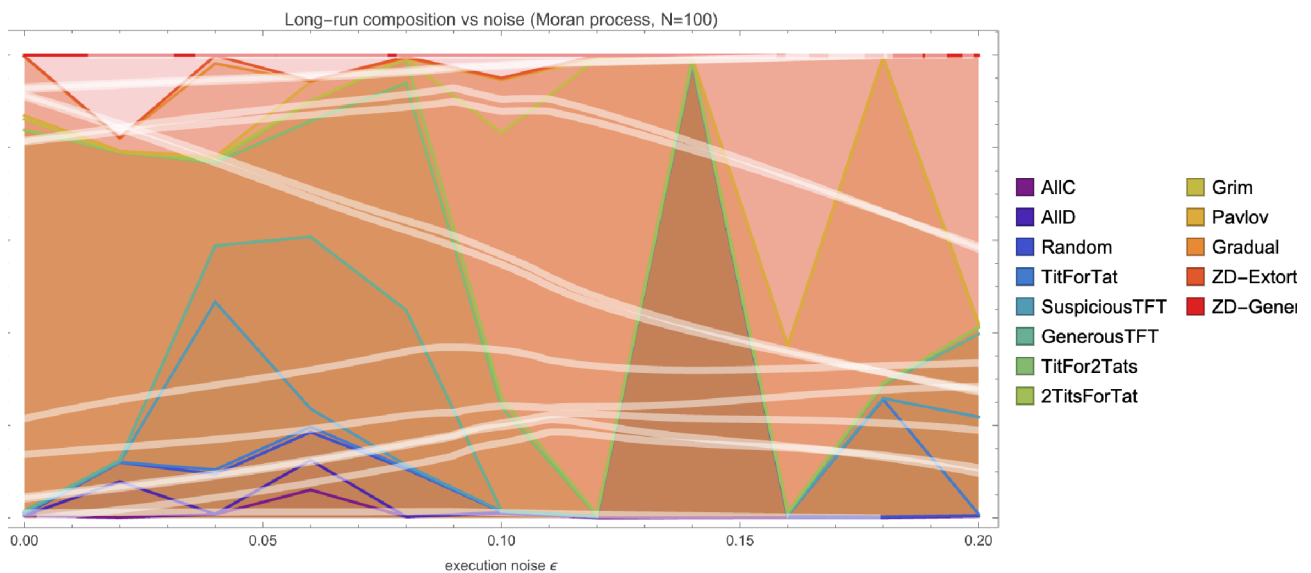
```
replicator[A_, steps_, mut_] := Module[{x = ConstantArray[1./n, n], f, phi},
  Do[ f = A . x; phi = x . f;
    x = x f / phi; (* selection *)
    x = (1 - mut) x + mut/n; (* small mutation term *)
    x = x / Total[x],
    {steps}];
  x]
```

Sweeping the noise level and recording the stable composition gives the headline figure (top of this post). It reveals **three regimes**:

- **Low noise ($\epsilon \approx 0$):** a coexistence of cooperative strategies – Generous TFT, Tit-for-2-Tats, ZD-Generous, Tit-for-Tat and Pavlov share the population. Cooperation is evolutionarily robust.
- **Intermediate noise:** the forgiving reciprocators (Generous TFT, Tit-for-2-Tats, ZD-Generous) dominate and trade the lead among themselves – error-correction is at a premium.
- **High noise ($\epsilon \geq 0.2$):** cooperation collapses. AllD sweeps to ~99% of the population: under heavy noise, reciprocators spend so long accidentally punishing each other that unconditional defection becomes the only stable attractor.

A finite-population check: the Moran process

Replicator dynamics is a deterministic, infinite-population limit. To confirm the result is not an artefact of that idealisation, we also run a stochastic **Moran process** (population $N=100$, frequency-dependent birth-death with rare mutation) and record the long-run average composition.



Moran process, long-run composition vs noise. Finite-population stochasticity makes the picture grainier, but the qualitative story survives: forgiving cooperators at low-to-moderate noise, defection-dominated outcomes as noise grows.

Takeaway. "Which strategy is evolutionarily stable?" has *no noise-independent answer*. The same population that converges on generous reciprocity in a clean channel converges on pure defection in a noisy one.

4. Discovering strategies that beat Tit-for-Tat under noise

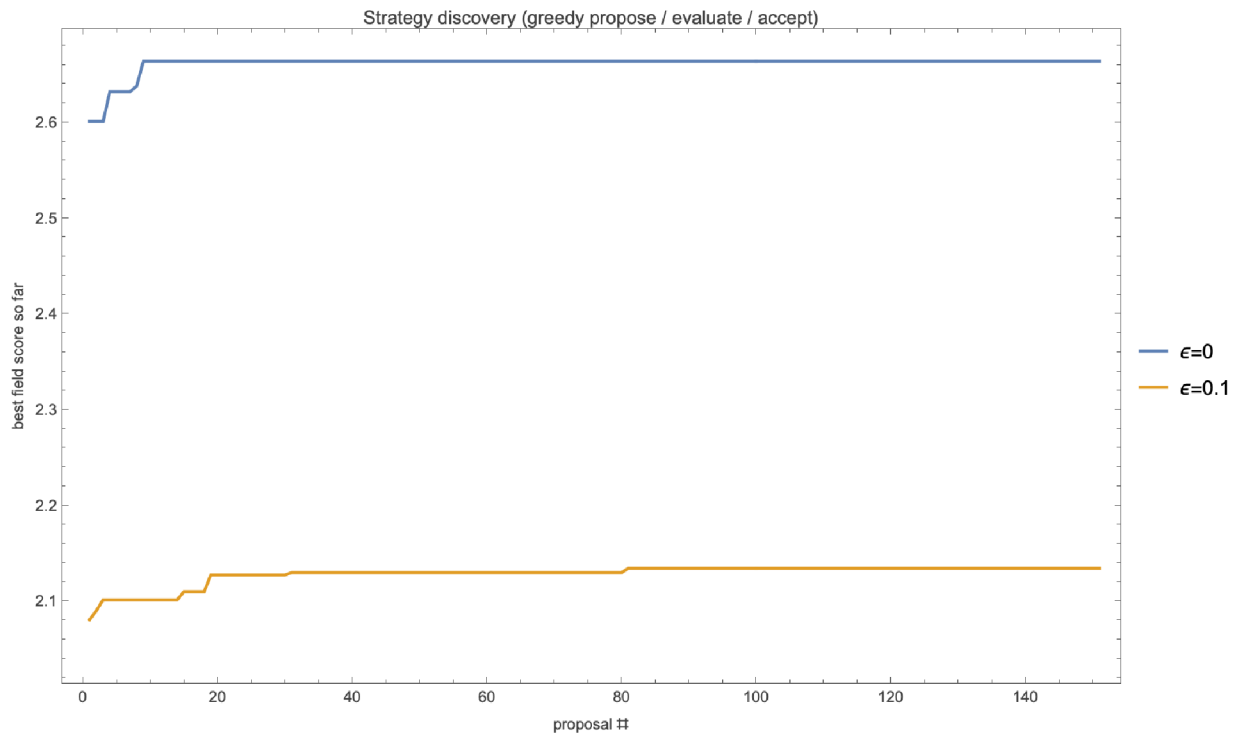
Can we do better than the textbook strategies? In the spirit of an automated-research loop – *propose* → *evaluate* → *accept* → *repeat* – we search the memory-one space $\theta = (open, pCC, pCD, pDC, pDD) \in [0,1]^5$ for the best response to the standard field at each noise level. A greedy hill-climb gives the narrative; a small genetic algorithm is the workhorse.

```

fitness[theta_, eps_] := Mean[
  meanMatch[memOne[theta[[1]], theta[[2;;]]], #, rounds, eps, reps][[1]] & /@ field];

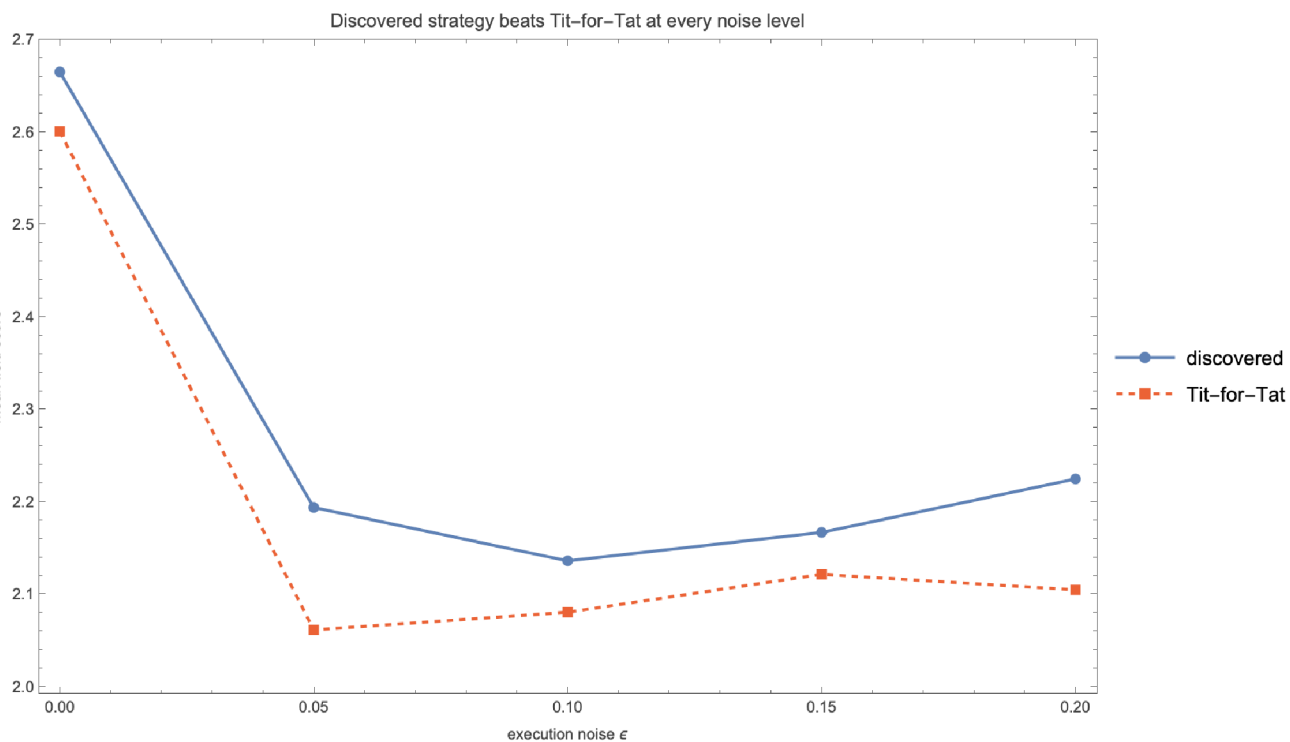
(* greedy accept/reject loop -- the autoresearch analogy *)
champion = {1, 1, 0, 1, 0}; (* start from Tit-for-Tat *)
Do[ proposal = clip[champion + RandomVariate[NormalDistribution[0, .12], 5]];
  If[fitness[proposal, eps] > fitness[champion, eps], champion = proposal],
  {150}]

```



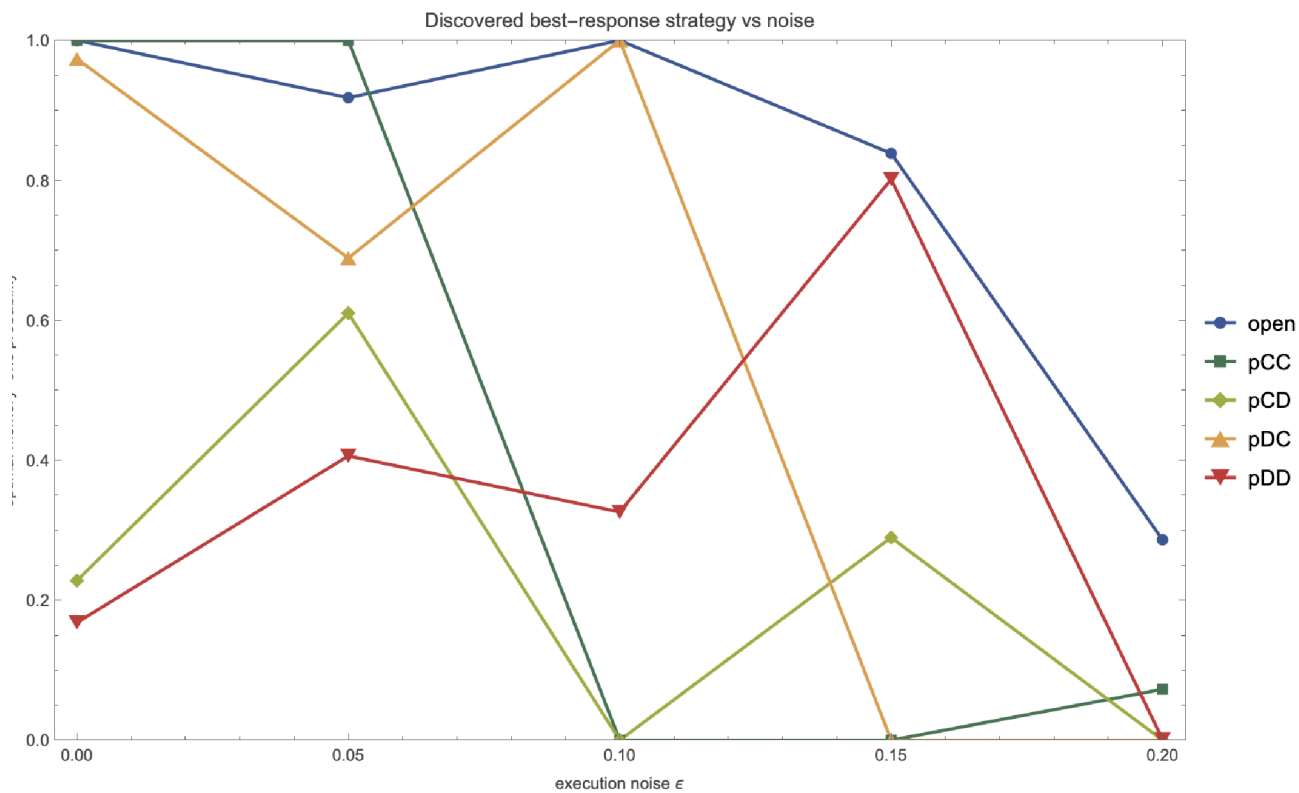
Greedy discovery: best field score found so far, starting from Tit-for-Tat, at two noise levels. Each upward step is an accepted proposal.

The discovered strategies beat Tit-for-Tat at every noise level



Discovered best-response strategy (solid) vs plain Tit-for-Tat (dashed) in field score, across noise. The discovered strategy wins everywhere; the margin is largest where noise matters most.

And the *shape* of the optimal strategy drifts with noise:



The discovered memory-one probabilities vs noise. The forgiveness parameter pCD (cooperate again after being suckered) rises with noise – error-correction becomes essential – until, at very high noise, the best response against a decaying field tilts back toward defection.

This is the same lesson as the evolutionary tournament, reached by a different route: **the optimal behaviour is a function of the environment**, and noise is the knob that turns generous reciprocity into defection.

5. Conclusions

- **Tit-for-Tat is excellent but not optimal.** In a cooperative field, more forgiving reciprocators (Generous TFT, Tit-for-2-Tats) edge it out – and extortion finishes last.
- **Evolutionary stability is noise-dependent.** Replicator and Moran dynamics agree: generous cooperation is stable at low noise; unconditional defection takes over at high noise.
- **Forgiveness is the answer to noise.** Both the evolutionary winners and the algorithmically discovered strategies become more forgiving as noise rises – up to the point where the whole field collapses.

Reproducing this work

Everything here is a few seconds to a few minutes of Wolfram Language – no external data, no API keys. The full source is on GitHub.

```
wolframscript -file tests/sanity.wls      # correctness checks
wolframscript -file wolfram/tournament.wls # round-robin ranking
wolframscript -file wolfram/evolution.wls  # replicator + Moran vs noise
wolframscript -file wolfram/discover.wls   # strategy discovery
wolframscript -file wolfram/figures.wls    # regenerate all figures
```

Built entirely in the Wolfram Language. The strategy-discovery loop borrows the propose → evaluate → accept shape of automated-research scaffolds, re-implemented natively in WL.